Rolling-window U.S. stock returns

$100 annual DCA

Monthly data, Aug 1926 – Mar 2026 (1,195 months)

Generated 2026-05-16

Full-history compound annual growth rate (market)

Nominal	10.20%	[1]
Real (CPI-deflated)	6.99%	[1][2]

Mean 10-year rolling IRR

Nominal	11.06%	[3]
Real	7.43%	[3]

Highest 10-year rolling IRR

Nominal	+20.90%	[3]
Real	+17.99%	[3]

Lowest 10-year rolling IRR

Nominal	-6.17%	[3]
Real	-11.31%	[3]

10-year rolling windows with IRR ≤ 0

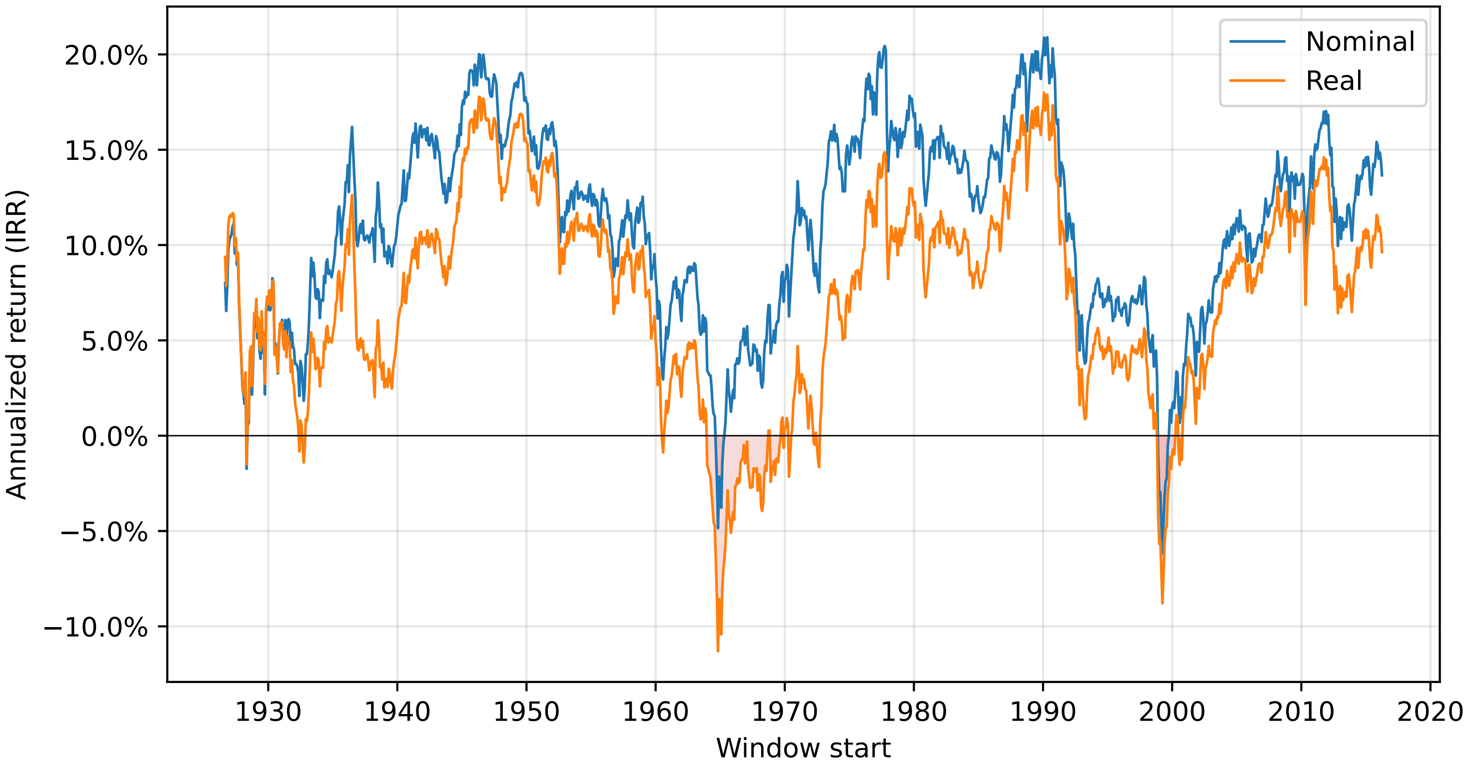
Nominal	1.86% (20 of 1,076)	[3]
Real	9.67% (104 of 1,076)	[3]

Total 10-year rolling windows	1,076	[3]
Window labeling	by start date	[3]
Total contributed per window	$1,000	

Invest $100 on each yearly anniversary of the window start — ten deposits over ten years, $1,000 total contributed. The metric is the money-weighted internal rate of return (IRR) — the annual rate that discounts the ten deposits to the terminal value. Comparing IRR here to CAGR in the lump-sum scenario shows how much the timing of deployed capital matters.

10-year rolling IRR — Nominal & Real

Each point is the annualized return of a 10-year window beginning on that month. Red shading marks the negative region.

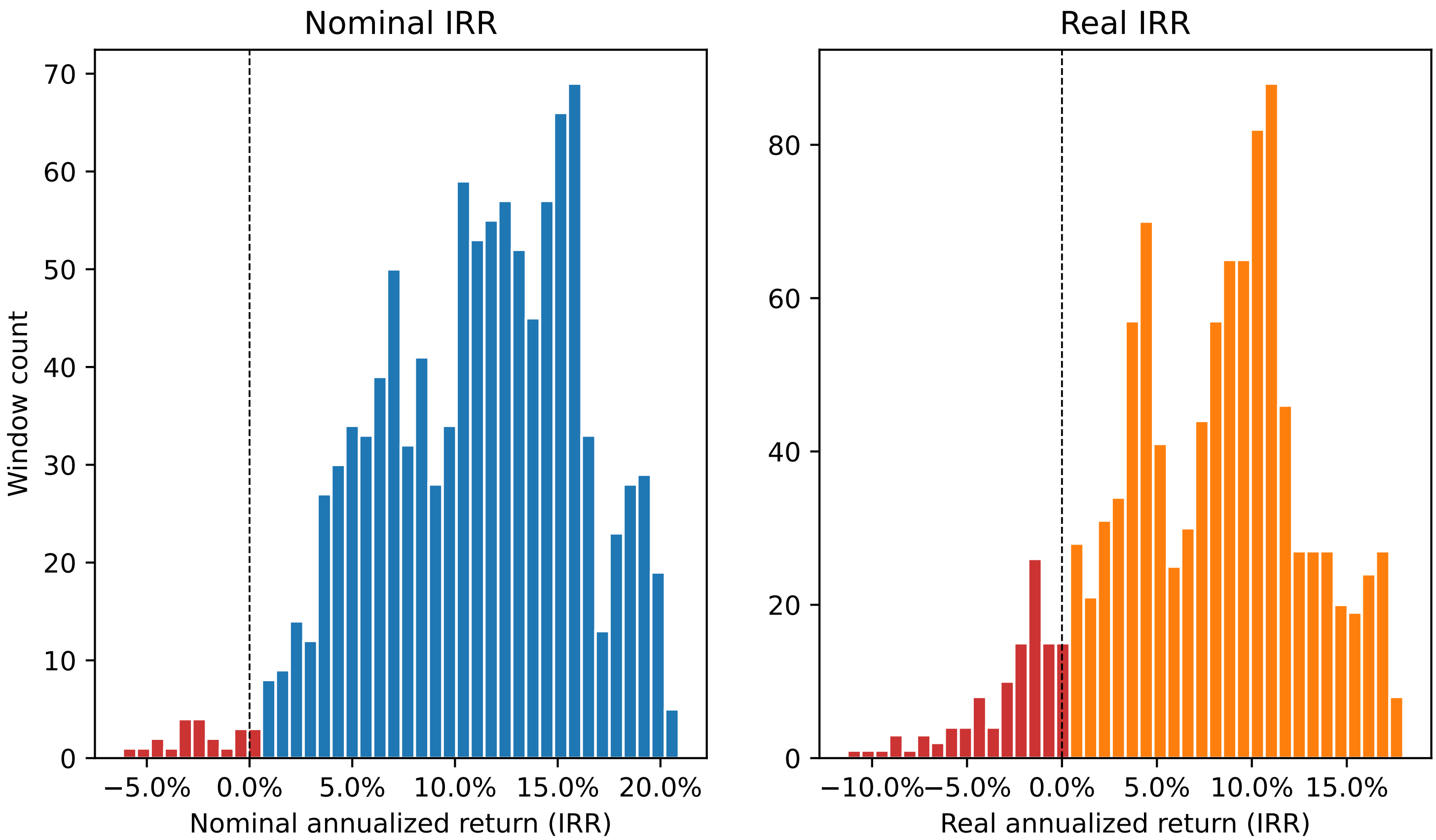


Summary statistics

	Nominal	Real
Mean	11.06%	7.43%
Median	11.59%	8.34%
Std. dev.	5.01%	5.33%
Min	-6.17%	-11.31%
Max	20.90%	17.99%
Share ≤ 0	1.86%	9.67%

Distribution of 10-year rolling IRR — Nominal & Real

Histogram of every 10-year window in the sample. Negative bins are shown in red.



Share of windows with IRR ≤ 0

Nominal: 1.86% (20 of 1,076 windows)

Real: 9.67% (104 of 1,076 windows)

“≤ 0” means an investor who followed this schedule over the full 10-year window would have ended at or below their total contributions, after compounding (and after inflation, for the real series).

[3]

Best and worst 10-year windows

Top 5 highest and lowest annualized returns (IRR), ranked by window-start month.

Best nominal

Start	End	IRR	$1,000 →
Apr 1990	Mar 2000	20.90%	$3,281
Jan 1990	Dec 1999	20.89%	$3,278
Sep 1977	Aug 1987	20.45%	$3,196
Sep 1990	Aug 2000	20.33%	$3,174
Aug 1977	Jul 1987	20.30%	$3,169

Worst nominal

Start	End	IRR	$1,000 →
Mar 1999	Feb 2009	**-6.17%**	**$716**
Oct 1964	Sep 1974	**-4.85%**	**$768**
Apr 1999	Mar 2009	**-4.67%**	**$776**
Feb 1999	Jan 2009	**-4.49%**	**$784**
Jan 1965	Dec 1974	**-3.78%**	**$814**

Best real

Start	End	IRR	$1,000 →
Jan 1990	Dec 1999	17.99%	$2,774
Apr 1990	Mar 2000	17.89%	$2,758
Apr 1946	Mar 1956	17.78%	$2,740
May 1946	Apr 1956	17.74%	$2,734
Aug 1946	Jul 1956	17.69%	$2,726

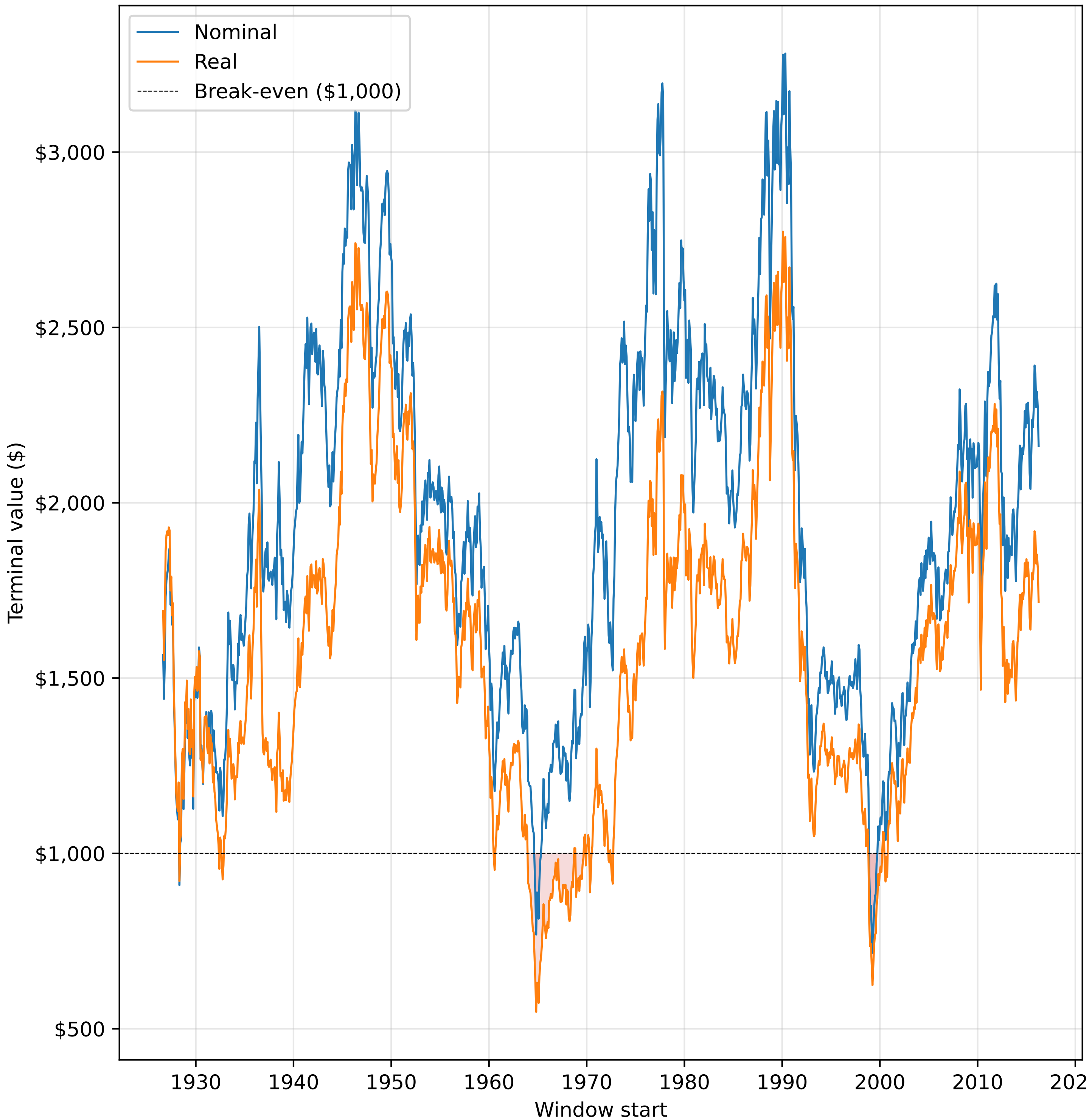
Worst real

Start	End	IRR	$1,000 →
Oct 1964	Sep 1974	**-11.31%**	**$548**
Jan 1965	Dec 1974	**-10.43%**	**$573**
Dec 1964	Nov 1974	**-9.46%**	**$603**
Mar 1999	Feb 2009	**-8.79%**	**$624**
Nov 1964	Oct 1974	**-8.57%**	**$631**

minal value of the full contribution schedule, held to the end of the 10-year window. [4] Red marks a negative IRR or a terminal below th

Terminal value of $1,000 after 10 years — Nominal & Real

re $1,000 contributed under this schedule would have landed, by start month. Red shading marks where the terminal fell below break-e



Sources and methodology

Data sources

[1] Fama/French Research Data Factors, monthly file. Ken French Data Library,
Tuck School of Business, Dartmouth.
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
Columns used: Mkt-RF (market excess return), RF (risk-free rate).
Values are published as percent (2.50 = 2.5%) and divided by 100 on ingest.

[2] U.S. CPI for All Urban Consumers, not seasonally adjusted (CPIAUCNS).
Federal Reserve Bank of St. Louis (FRED).
https://fred.stlouisfed.org/series/CPIAUCNS
The not-seasonally-adjusted series is used deliberately – the SA series
(CPIAUCSL) only begins in 1947 and would truncate 21 years of history.

Series construction

```
nominal_return = (Mkt-RF + RF) / 100  – total U.S. market return.
inflation      = CPI.pct_change()    – month-over-month CPI change.
real_return    = (1 + nominal) / (1 + inflation) − 1   (Fisher equation).
```

The first observation (Jul 1926) is dropped because monthly inflation
requires a prior CPI reading.

[3] Rolling 10-year window math — $100 annual DCA

For each window start t, deposit $100 at months t+0, t+12, ..., t+108
(ten yearly anniversaries). Each deposit at month m grows by the product
of monthly returns from m to the end of the window (month t+120):

```
  grown_k     = $100 * exp( Σ log(1 + r_i) )   for i in [t+12k, t+120-1]
  terminal(t) = sum of grown_0 ... grown_9
```

IRR(t) is the annual rate R solving:

```
  Σ $100 * (1+R) ^ (10-k) = terminal(t)   for k in 0..9
```

Solved by bisection. Same start-month labeling and tail-drop rules as the
lump-sum scenario.

[4] Full-history market CAGR

```
terminal = exp( Σ log(1 + r_t) ) over the full sample of monthly returns.
CAGR     = terminal ^ (12 / n_months) − 1.
```

Shown on the cover for context; independent of the scenario schedule.

Reproducibility

```
python -m src.ingest      # rebuild data/processed/monthly_returns.parquet
python -m src.export      # rebuild docs/data/*.json
python -m src.report      # rebuild docs/reports/*.pdf
```

Sample window: Aug 1926 – Mar 2026 (1,195 monthly observations).